



8/7/2026

ASSIGNMENT DAY3

Basic Network Configuration.

TASK 1.

- ❖ Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

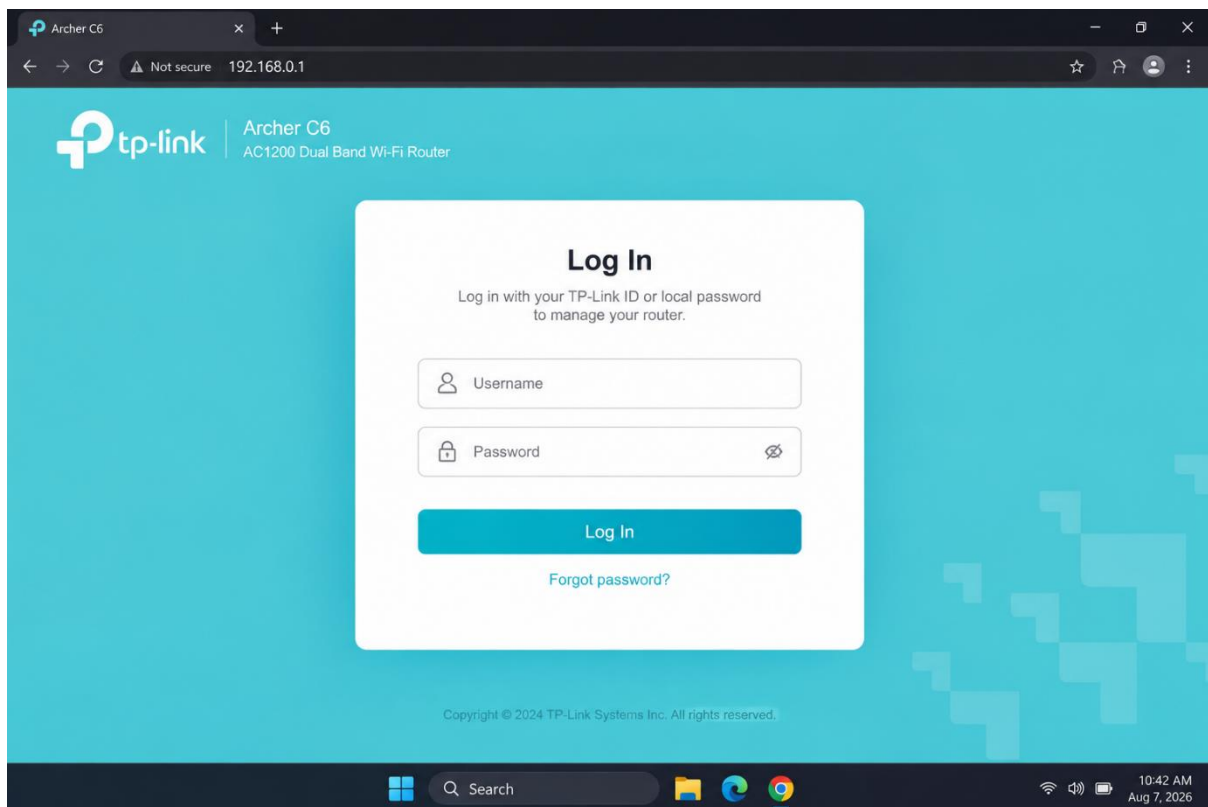


Figure : Router Admin Login Page Using the Default IP Address (192.168.0.1).

Explanation :

- ✓ The image shows the router admin login page opened using the default IP address 192.168.0.1. It displays the Username and Password fields required to access the router's settings. This login page is used to configure and manage the router.

TASK 2.

- ❖ Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.



```
C:\WINDOWS\system32\cmd. X + -
Connection-specific DNS Suffix . : 
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 00-50-56-C8-00-08
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::bdd:1834:ae9:6264%13(Preferred)
IPv4 Address. . . . . : 192.168.148.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 07 August 2026 09:18:51
Lease Expires . . . . . : 07 August 2026 10:49:49
Default Gateway . . . . . : 
DHCP Server . . . . . : 192.168.148.254
DHCPv6 IAID . . . . . : 704663638
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-75-A3-80-08-BF-B8-C7-81-A9
Primary WINS Server . . . . . : 192.168.148.2
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : C8-5E-A9-F7-F8-91
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::25a0:6620:f277:92c0%5(Preferred)
IPv4 Address. . . . . : 192.168.167.101(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 07 August 2026 09:18:54
Lease Expires . . . . . : 07 August 2026 12:19:01
Default Gateway . . . . . : 192.168.167.1
DHCP Server . . . . . : 192.168.167.1
DHCPv6 IAID . . . . . : 80240297
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-75-A3-80-08-BF-B8-C7-81-A9
DNS Servers . . . . . : 192.168.167.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\callm>
```

Figure : Network Configuration Details (IP Address, Subnet Mask, and Default Gateway).

Network Details :

<u>Network Details</u>	<u>Value</u>
IP Address (IPv4)	192.168.167.101
Subnet Mask	255.255.255.0
Default Gateway	192.168.167.1

Explanation :

- ✓ The screenshot shows the network configuration of my laptop connected to a Wi-Fi network. The IP address, subnet mask, and default gateway were obtained using the ipconfig command in Command Prompt.

TASK 3.

- ❖ **Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.**
Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

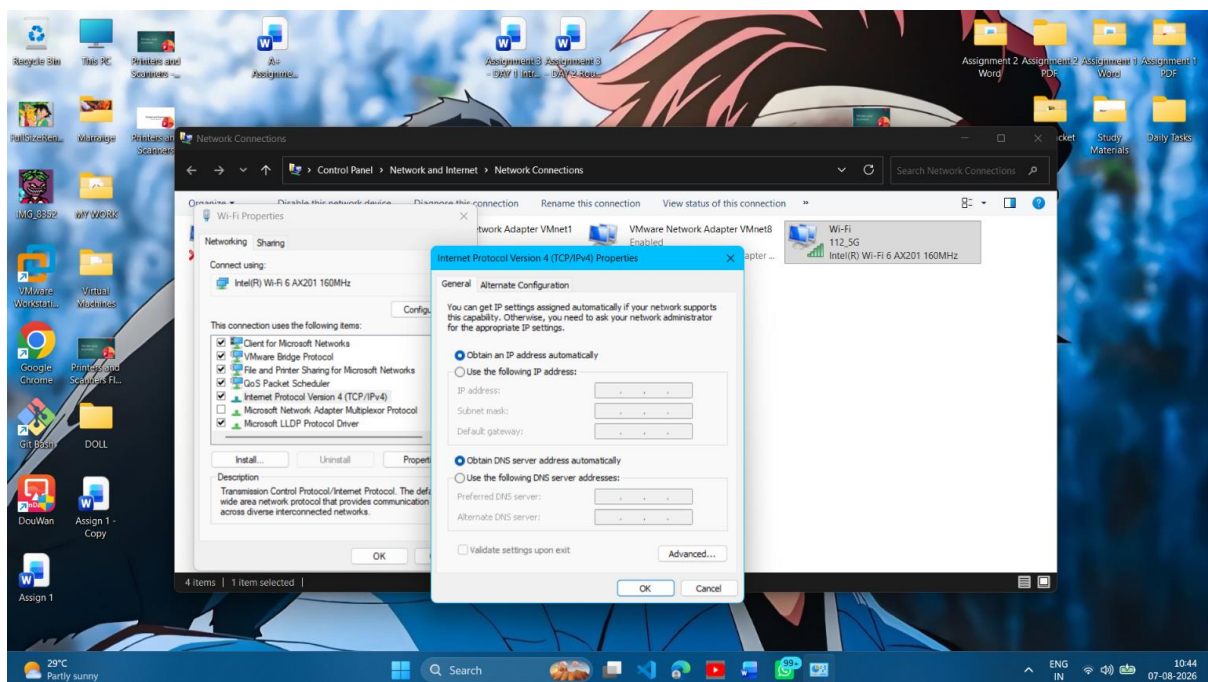


Figure 1 : DHCP (Obtain IP Automatically) Configuration.

Explanation :

- ✓ The device was initially configured to Obtain an IP address automatically (DHCP). This allows the router to assign the network settings automatically.

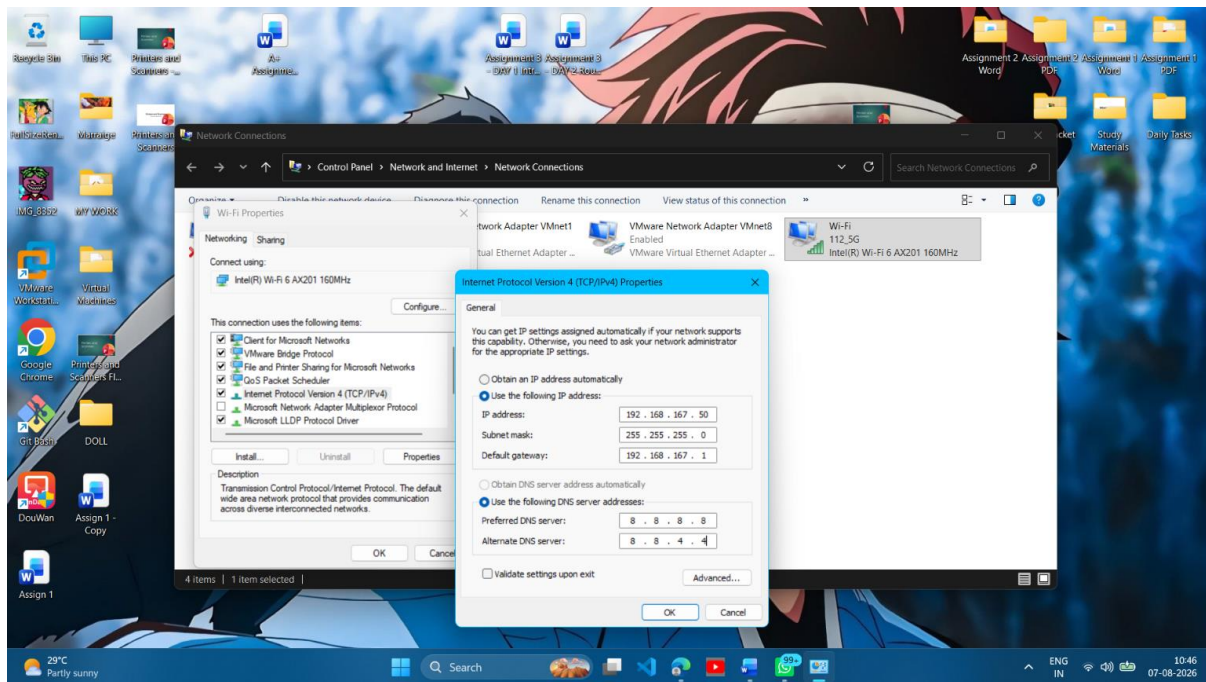


Figure 2 : Static IP Address Configuration.

Explanation :

- ✓ The IP settings were changed to a static IP address in the same network as the router. The following values were used:

- ✚ IP Address: 192.168.167.50
- ✚ Subnet Mask: 255.255.255.0
- ✚ Default Gateway: 192.168.167.1

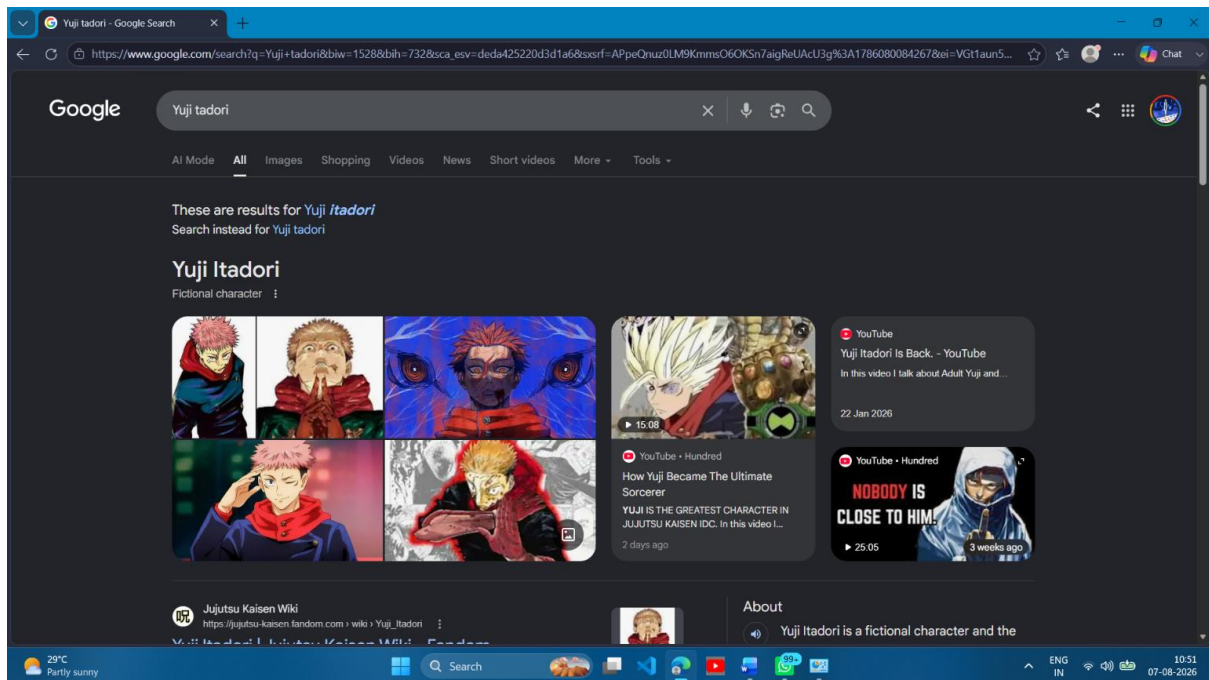


Figure: Internet Connection Test.

Explanation :

- ✓ After setting the static IP address, the Internet connection was tested by opening a website. The website opened successfully, confirming that the static IP configuration was working correctly.

Result :

- ✓ The static IP address was configured successfully, and the Internet connection worked properly. After testing, the settings were changed back to Obtain IP automatically (DHCP) to avoid network issues.

TASK 4.

- ❖ In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

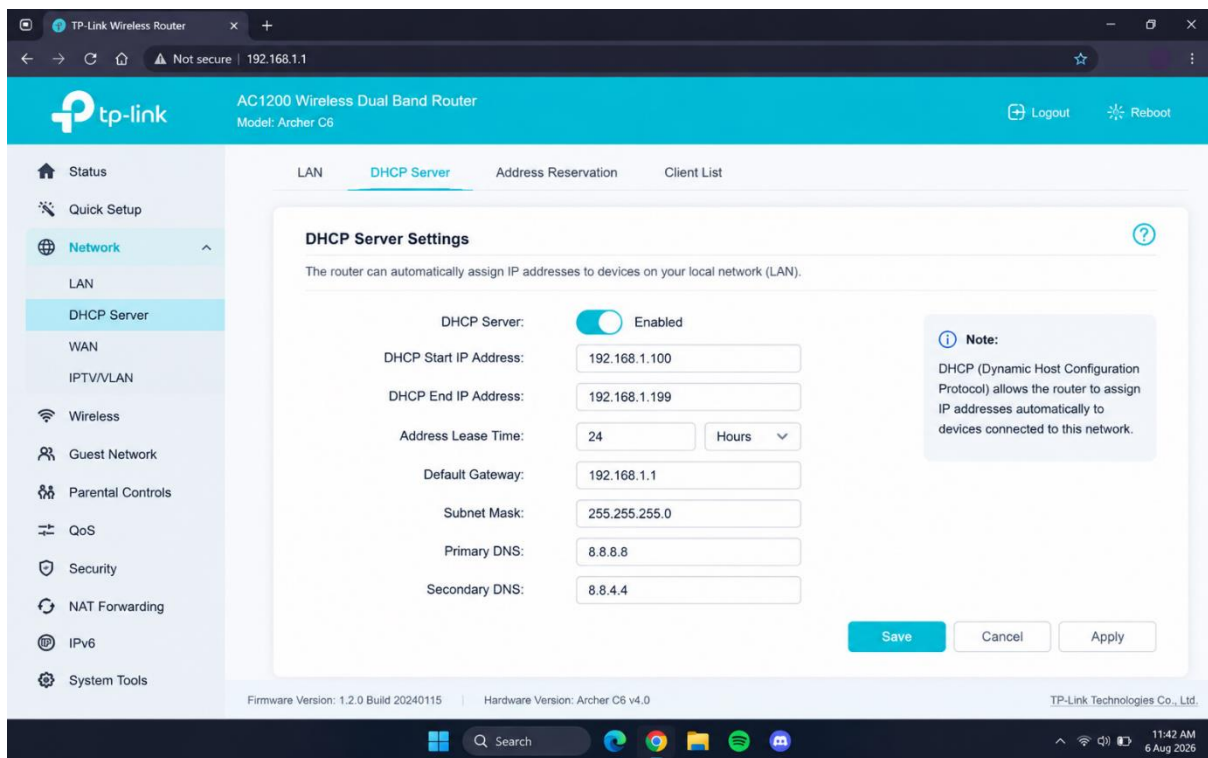


Figure : DHCP Server Configuration Showing the Current DHCP Range.

✚ **Current DHCP Range: 192.168.1.100 – 192.168.1.199**

Explanation :

- ✓ The DHCP server automatically assigns IP addresses to devices connected to the network. If two devices are assigned the same IP address, an **IP conflict** occurs. As a result, one or both devices may lose internet access or experience network connection problems because the router cannot identify which device should receive the data.